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(2) 令 $u = U(x,p) = \int_0^{\infty} u(x,t) e^{-pt} dt$, $\frac{du}{dt} = pU(x,p)$.

$\Rightarrow pU = k\frac{d^2U}{dx^2} \Rightarrow U = c_1 e^{\sqrt{\frac{p}{k}}x} + c_2 e^{-\sqrt{\frac{p}{k}}x}$

因边值 $U = \frac{U_0}{p} e^{-\sqrt{\frac{p}{k}}x}$

查表得:

$$u = u_0 \frac{1}{p} e^{-\sqrt{\frac{p}{k}}x} \cdot 1$$

$$= u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{kt}}\right)$$

4、拉普拉斯变换

$p^2 U(x,p) - a^2 \frac{d^2 U(x,p)}{dx^2} = F(x,p) + p\phi(x) + \psi(x)$

常数变易

$$U(x,p) = \frac{1}{2a} \int_{-\infty}^{\infty} \left[\phi(x') + \frac{\psi(x')}{p} + \frac{F(x,p)}{p}\right] e^{-\frac{p}{a}|x-x'|} dx'$$

又 $\int_{-\infty}^{+\infty} F(x,p) \cdot \frac{e^{-\frac{p}{a}(x-x')}}{p} dx = \int_0^t \int_{-\infty}^{\infty} f(x';\tau) \cdot \eta\left(t - \frac{x-x'}{a} - \tau\right) dx' d\tau = \int_0^t \frac{1}{2a} \int_{x-ct+c\tau}^{x+at-c\tau} f(x';\tau) dx'$

$\Rightarrow U(x,p) = \frac{1}{2}\left[\phi(x-at) + \phi(x+at)\right] + \frac{1}{2a}\left[\int_0^t \int_{x-at+a\tau}^{x+at-a\tau} f(x';\tau)\, dx' d\tau + \int_{x-ct}^{x+at} \psi(x') dx'\right]$

5、升维为三维柱面

$$\begin{cases} \frac{d^2 U}{dt^2} + a^2 k^2 U = 0 \\ U|_{t=0} = \Phi(\vec{k}),\ \frac{du}{dt} = \Psi(\vec{k}) \text{，由前角为} \end{cases}$$

$U(t, k_x, k_y) = \frac{\Psi(\vec{k})\sin(akt)}{ak} + \Phi(\vec{k})\cos(akt)$

又原函数 $= \int_0^{\infty}\int_0^{\pi}\int_0^{2\pi} \frac{\sin(akt)}{ak} e^{i\vec{k}\cdot\vec{r}} \cdot k^2 \sin\theta\, d\varphi\, d\theta\, dk \cdot \frac{1}{(\sqrt{2\pi})^3}$

$$= \frac{1}{\sqrt{2\pi}} \int_0^\infty \frac{\sin(a|t|)\left[e^{ikr} - e^{-ikr}\right]}{ikr} dk$$

$$= \frac{\sqrt{2\pi}\,\delta(r-at)}{2ar}$$

第一项卷积，原函数

$$= \frac{1}{4\pi a} \iiint \frac{\psi(x',y')\,\delta\left(\sqrt{(x-x')^2+(y-y')^2+z'^2} - at\right)}{\sqrt{(x-x')^2+(y-y')^2+z'^2}} dx'dy'dz'$$

令 $z=0$，又第二项为第一项的导数，得：

$$u(x,y) = \frac{1}{2\pi a}\left(1+\frac{\partial}{\partial t}\right) \iint \frac{\psi(x',y')}{\sqrt{a^2t^2-(x-x')^2-(y-y')^2}} dx'dy'$$